卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-TM-0502 30-B003701S-D0502

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图纸文件目录(首页)
CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

工程 默蒂亚里±660kV换流站 ±660kV Matiari Converter Station PROJECT
阶段 施工图 Detail Design DESIGN STAGE
专业 建筑机电 TM SPECIALITY
卷 第 5 卷 VOLUME 5
册 第 02 册 BOOK 02
卷册 名称 阀厅建筑电气图 Volume name Valve hall Electrical Drawings

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY / SURVEYED BY	
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序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-D0502-001	卷册说明 Volume instructions	A	1		
2	30-B003701S-D0502-002	极2阀厅底层照明布置图 Lighting plan of ground floor of pole 2 valve hall	A	1		
3	30-B003701S-D0502-003	极1阀厅底层照明布置图 Lighting plan of ground floor of pole 1 valve hall	A	1		
4	30-B003701S-D0502-004	极2阀厅巡视走道层照明布置图 Lighting plan of Patrol aisle floor of pole 2 valve hall	A	1		
5	30-B003701S-D0502-005	极1阀厅巡视走道层照明布置图 Lighting plan of Patrol aisle floor of pole 1 valve hall	A	1		
6	30-B003701S-D0502-006	极2阀厅屋架线槽布置图 Trunking plan on roof truss of pole 2 valve hall	A	1		
7	30-B003701S-D0502-007	极1阀厅屋架线槽布置图 Trunking plan on roof truss of pole 1 valve hall	A	1		
8	30-B003701S-D0502-008	插座箱及照明箱配置接线图 Distribution diagram of socket box and lighting box	A	1		
9	30-B003701S-D0502-009	材料表 Material list	A	1		

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卷册说明及图例

Volume instructions and legend

一、项目概况：

First: Project overview

项目名称：巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程/默蒂亚里±660kV换流站

Project name: ±660kV HVDC Project from Matiari to Lahore in Pakistan

建设单位：国家电网/中国电力技术装备有限公司

Development organization: State Grid / China Electric Power Equipment And Technology CO.,LTD

本次出图范围为:阀厅建筑电气图（包括极1、极2两个阀厅）

Design scope:valve hall electrical drawings(include pole 1 valve hall and pole 1 valve hall)

二、设计内容

Second: Design content

1. 本工程设计内容包括:阀厅底层照明布置和阀厅巡视走道层照明布置。

1. This design content including:lighting plan of ground floor and lighting plan of patrol aisle floor of valve hall.

三、供配电系统

Third: Design content

1. 极1、极2阀厅照明箱380/220V低压电源引自主控楼配电屏，极1、极2阀厅事故照明箱电源引自主控楼直流屏。极1阀厅照明箱、极2阀厅照明箱和事故照明箱均位于主控楼三楼。

1. The power supply of lighting distribution boxs in pole 1 valve hall and pole 2 valve hall come from distribution panel in control building,the power supply of DC-AC shifter in pole 1 valve hall and pole 2 valve hall come from DC cabinet in control building.Lighting distribution boxs in pole 1 valve hall and in pole 2 valve hall and DC-AC shifter are installed in control building.

四、导线选择与敷设

Fourth: Cable selection and installation

1. 导线选择

1. Cable selection

除图中已标注外，照明支线选用ZRB-YJV22-P-0.6/1kV-3×4屏蔽电缆,具体线型规格详见本卷册配电箱配置接线图。

Except marked in the drawing,the ZRB-YJV22-P-0.6/1kV-3×4 shielded cable is adopted as lighting cable,the specific type is shown in distribution diagram of distribution box.

2. 敷设方式

2. Installation

除图中已标注外，照明线路均穿热镀锌钢管保护。

Except marked in the drawing, the lighting and power cable are protected by the flame-retardant PVC pipe.

应急照明线路明敷时，应穿金属管或金属线槽，并且选用阻燃电缆。

When the emergency lighting cable is exposed in the wall,it should be flame-retardant cable and be laid in metal tube or metal trunking.

备用及疏散照明导线均采用耐火导线。

Both standby and evacuation lighting cable are fire-resistant cable.

五、灯具性能要求

Fifth: performance demands of luminaire

1. 荧光灯灯具的效率

1. The fluorescent luminaire lunminaire efficiency

灯具出光口形式 The emit form of the lunminaire	开敞式 Open type	保护罩（玻璃或塑料） Protection cover（glass or plastic）		格栅 Grille
		透明 Transparent	磨砂、棱镜 Frosted、prism	
灯具效率 Lunminaire efficiency	75%	70%	55%	65%

2. 发光二极管灯具的效率

2. The LED luminaire lunminaire efficiency

色温 colour temperature	2700K		3000K		4000K	
灯具出光口形式 The emit form of the lunminaire	格栅 Grille	保护罩 Protection cover	格栅 Grille	保护罩 Protection cover	格栅 Grille	保护罩 Protection cover
灯具效率 Lunminaire efficiency	55%	60%	55%	60%	55%	60%

3. 阀厅内照明采用荧光灯和LED高顶灯，所有荧光灯镇流器要求采用节能型电子式。照明光源的显色指数(Ra)不小于80,单灯功率因数不小于0.9。阀厅内所有灯具均不含紫外光。

3. The fluorescent luminaires and LED high bay luminaires are used in valve halls,energy-saving electronic ballast should be applied in all the fluorescent luminaires.the colour rendering index(Ra) of the luminaire source should be not less than 80,the single-luminaire power factor is not less than 0.9. All the luminaires in valve halls don't contain ultraviolet light.

六、其它

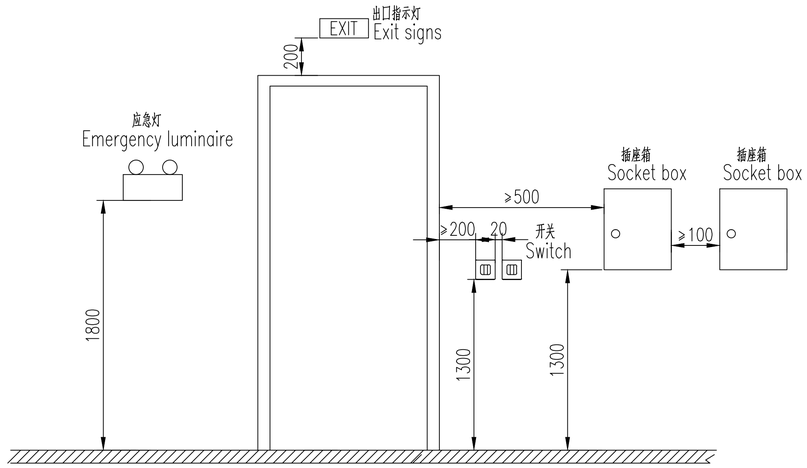
Sixth: Others

1. 其它未提之处均参照地方相关施工安装法规。

1. Other unmentioned places shall refer to local construction and installation regulations.

2. 本卷册设计以中文部分为准，英文部分仅做参考。

2. In this volume, the Chinese section should prevail, and the English section is for reference purposes only.



插座箱、开关及部分灯具安装高度示意图

Schematic of installation height of socket box,switch and some luminaires

巴基斯坦默蒂亚里—拉合尔直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

默蒂亚里±660kV换流站

±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - M C S - C S I - D D - T M - 0 5 0 2 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡如	2018.9.17
批准 Approved	[illegible]	2018.9.18

卷册说明

Volume instructions

比例 Scale:

图号 DWG NO.: 30-B003701S-D0502-001

版本 REV.: A

420X594

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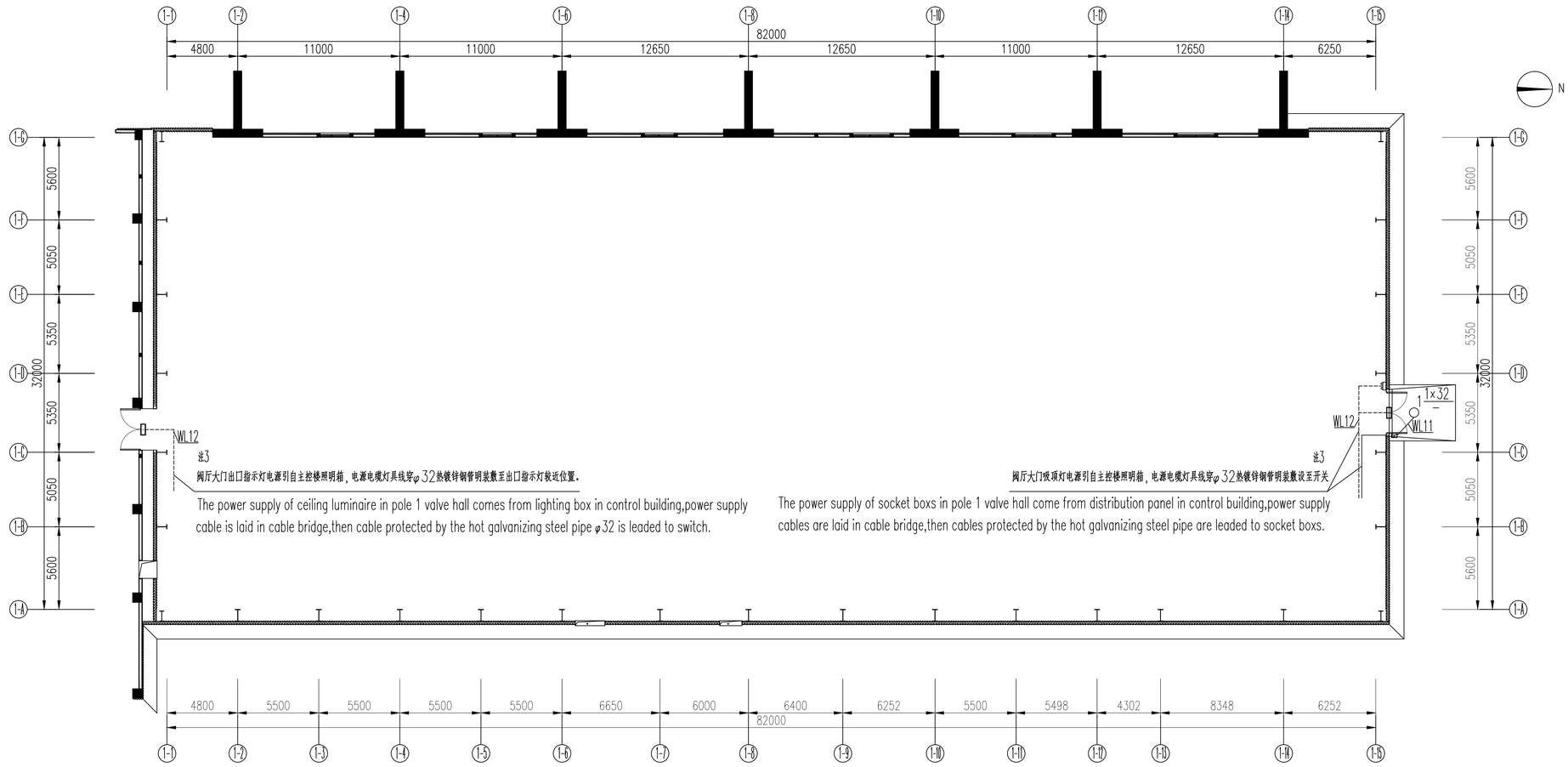
图形文件名



说明：

1. 极1阀厅大门吸顶灯电源引自主控楼照明箱，电源电缆灯具线穿φ32热镀锌钢管明装敷设至开关，材料工总于材料工总表。
2. 极1阀厅大门出口指示灯电源引自主控楼照明箱，电源电缆灯具线穿φ32热镀锌钢管明装敷至出口指示灯较近位置，材料工总于材料工总表。
3. 本略册灯具敷设原则采用穿管敷设，材料已工总于材料工总表，如施工方考虑便于施工等原因，可借用附近有余量的桥架等电缆通道。

Description:

1. The power supply of socket boxs in pole 1 valve hall come from distribution panel in control building,power supply cables are laid in cable bridge,then cables protected by the hot galvanizing steel pipe are leaded to socket boxs.
2. The power supply of ceiling luminaire in pole 1 valve hall comes from lighting box in control building,power supply cable is laid in cable bridge,then cable protected by the hot galvanizing steel pipe φ32 is leaded to switch.

材料表
Material list

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	图例 legend	备注 Notes
1	吸顶灯(防水防尘型) Ceiling luminaire (waterproof and dustproof)	AC220V 1×32W 光通量不小于2500lm AC220V 1×32W The luminous flux is not less than 2500 lm	套 Set	1	○	不含紫外光 Without ultraviolet light
2	出口指示灯 Exit signs	光源LED(自带蓄电池,放电时间不低于120分钟) AC220V 3W 光通量不小于300lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) the luminous flux is not less than 300 lm	套 Set	2	▭	不含紫外光 Without ultraviolet light
3	应急灯 Emergency luminaire	AC220V 2×8W 光通量不小于1600lm 带蓄电池,具放电时间不低于120分钟 AC220V 2×8W The luminous flux is not less than 1600 lm,the discharge time of battery is not less than 120min	套 Set	1		不含紫外光 Without ultraviolet light
4	单联单控开关(防尘防水型) Single board single control switch (waterproof and dustproof)	AC250V 10A	只 Pcs	1		
5	屏蔽电缆 Shielded cable	ZRB-YJV22-P-0.6/1kV-3×4	米 Meter	160		吸顶灯用 For ceiling luminaire
6	屏蔽电缆 Shielded cable	NH-YJV22-P-0.6/1kV-3×4	米 Meter	170		出口指示灯和应急灯用 For exit signs and emergency luminaire
7						

巴基斯坦默蒂亚里—拉合尔直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID
中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

默蒂亚里±660kV换流站
±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - M C S - C S I - D D - T M - 0 5 0 2 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡如	2018.9.17
批准 Approved	[illegible]	2018.9.18

极2阀厅底层照明布置图
Lighting plan of ground floor of pole 2 valve hall

比例 Scale: 1:250
图号 DWG NO.: 30-B003701S-D0502-002
版本 REV.: A

420X594

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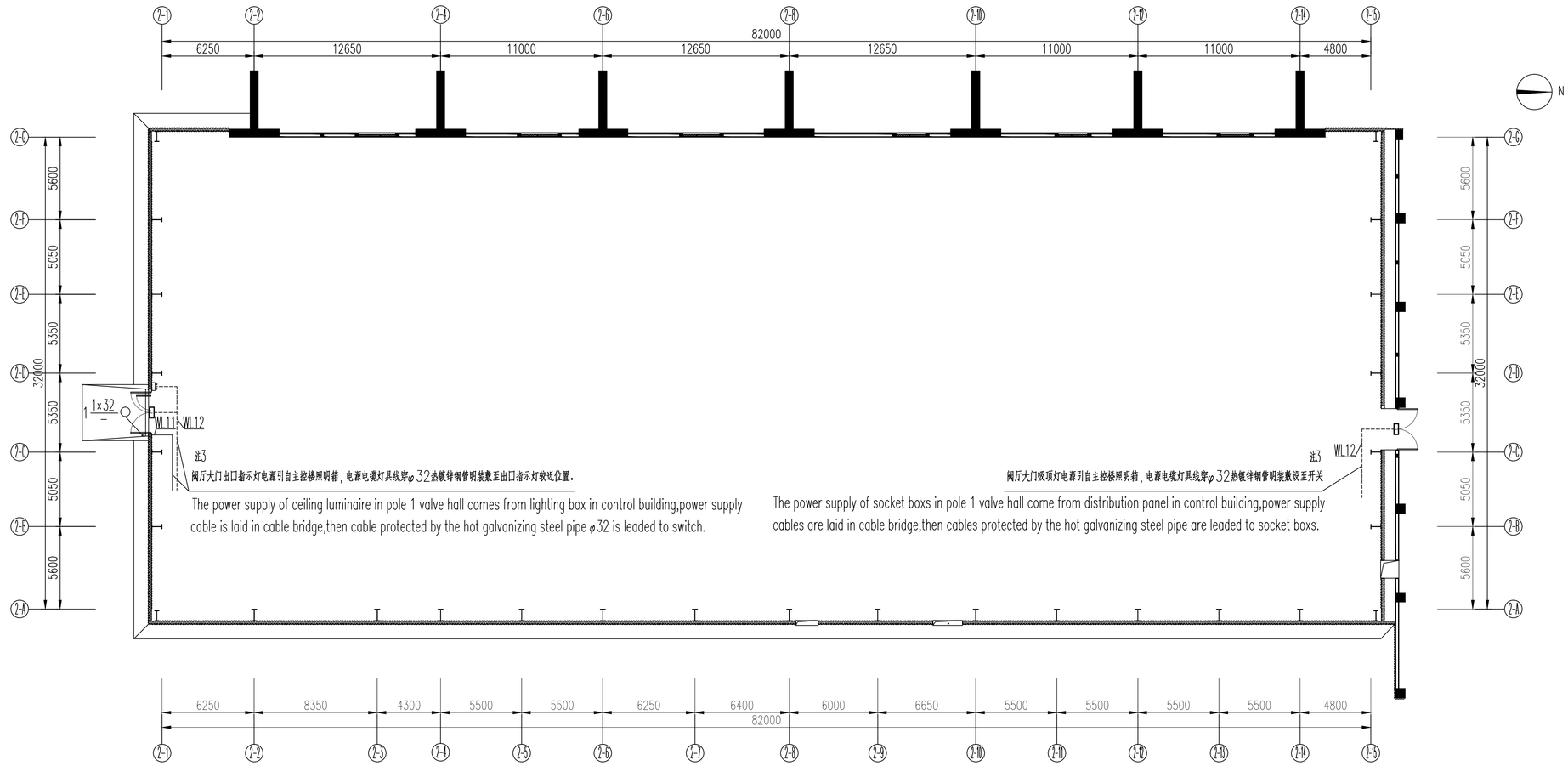

说明：

1. 极1阀厅大门吸顶灯电源引自主控楼照明箱，电源电缆沿电缆桥架敷设，至吸顶灯开关较近位置穿φ32热镀锌钢管至开关。
2. 极1阀厅大门出口指示灯电源引自主控楼照明箱，电源电缆沿电缆桥架敷设，至出口指示灯较近位置穿φ32热镀锌钢管至出口指示灯。
3. 本卷册灯具敷设原则采用穿管敷设，材料已汇总于材料汇总表，如施工方考虑便于施工等原因，可借用附近有余量的桥架等电缆通道。

Description:

1. The power supply of socket boxs in pole 1 valve hall come from distribution panel in control building,power supply cables are laid in cable bridge,then cables protected by the hot galvanizing steel pipe are leaded to socket boxs.
2. The power supply of ceiling luminaire in pole 1 valve hall comes from lighting box in control building,power supply cable is laid in cable bridge,then cable protected by the hot galvanizing steel pipe φ25 is leaded to switch.

材料表
Material list

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	图例 legend	备注 Notes
1	吸顶灯(防水防尘型) Ceiling luminaire (waterproof and dustproof)	AC220V 1×32W 光通量不小于2500lm AC220V 1×32W The luminous flux is not less than 2500 lm	套 Set	1	○	不含紫外光 Without ultraviolet light
2	出口指示灯 Exit signs	光源LED(自带蓄电池,放电时间不低于120分钟) AC220V 3W 光通量不小于300lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) The luminous flux is not less than 300 lm	套 Set	2	▭	不含紫外光 Without ultraviolet light
3	应急灯 Emergency luminaire	AC220V 2×8W 光通量不小于1600lm 带蓄电池,放电时间不低于120分钟 AC220V 2×8W The luminous flux is not less than 1600 lm,the discharge time of battery is not less than 120min	套 Set	1		不含紫外光 Without ultraviolet light
4	单联单控开关(防尘防水型) Single board single control switch (waterproof and dustproof)	AC250V 10A	只 Pcs	1		
5	屏蔽电缆 Shielded cable	ZRB-YJV22-P-0.6/1kV-3×4	米 Meter	160		吸顶灯用 For ceiling luminaire
6	屏蔽电缆 Shielded cable	NH-YJV22-P-0.6/1kV-3×4	米 Meter	170		出口指示灯和应急灯用 For exit signs and emergency luminaire
7						

巴基斯坦默蒂亚里—拉合尔直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

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±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - M C S - C S I - D D - T M - 0 5 0 2 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡如	2018.9.17
批准 Approved	[illegible]	2018.9.18

极1阀厅底层照明布置图
Lighting plan of ground floor of pole 1 valve hall

比例 Scale: 1:250
图号 DWG NO.: 30-B003701S-D0502-003
版本 REV.: A

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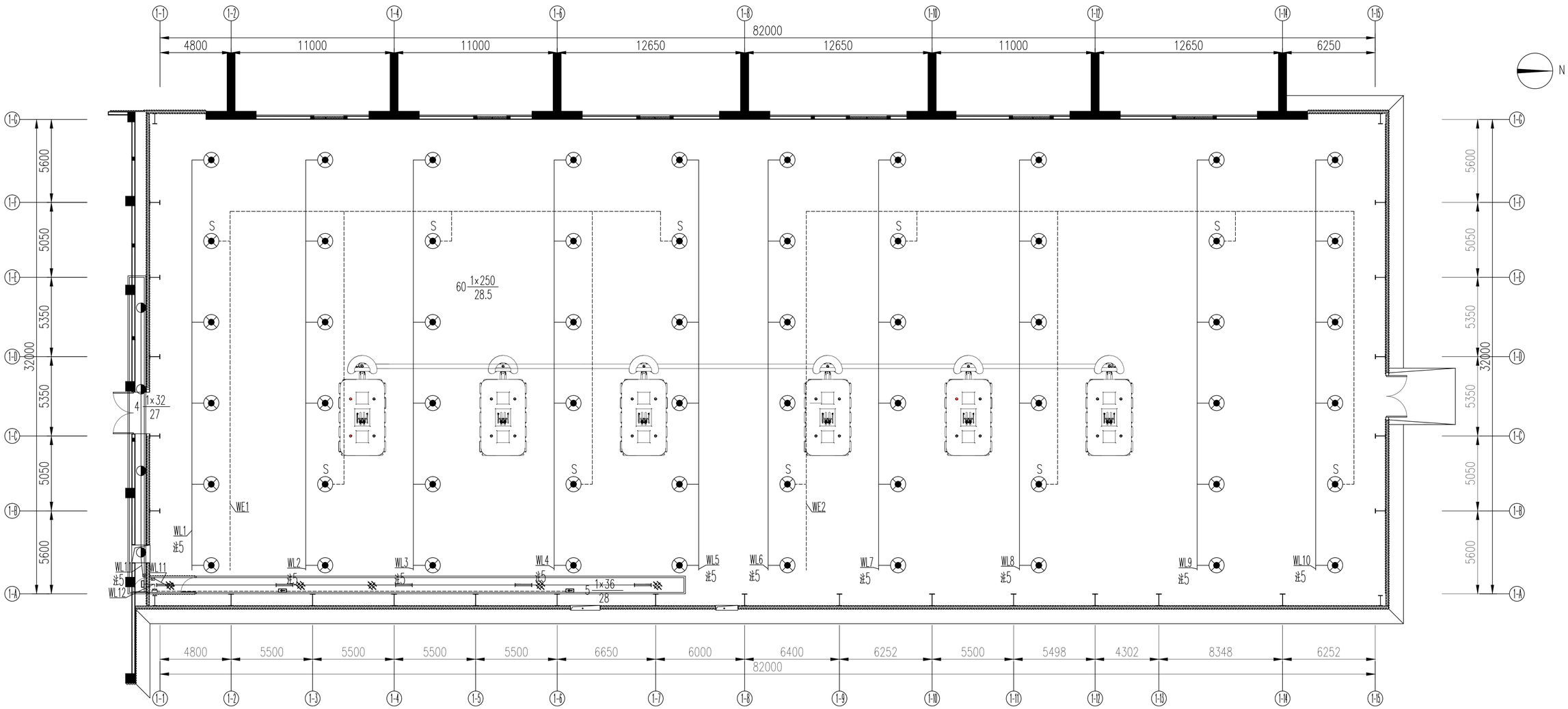
图形文件名



说明：

1. 极1阀厅正常照明高顶灯电源引自主控楼三楼照明箱，灯具线穿φ32热镀锌钢管明装敷设，至高顶灯就近位置经50×25线槽引至高顶灯，材料工总于材料工总表。
2. 极1阀厅备用照明高顶灯电源引自主控楼三楼事故照明箱，灯具线穿φ32热镀锌钢管明装敷设，至高顶灯就近位置经50×25线槽引至高顶灯，材料工总于材料工总表。
3. 极1阀厅壁灯和荧光灯电源引自主控楼三楼照明箱，灯具线穿φ32热镀锌钢管明装敷设引至灯具，材料工总于材料工总表。
4. 极1阀厅巡视走道出口指示灯和应急灯电源引自主控楼三楼照明箱，灯具线穿φ32热镀锌钢管明装敷设热镀锌钢管引至灯具，材料工总于材料工总表。
5. 本表带灯具敷设原则采用穿管敷设，材料已工总于材料工总表，如施工方考虑便于施工等原因，可借用附近有余量的桥架等电缆通道。

材料表
Material list

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	图例 legend	备注 Notes
1	高顶灯 High bay luminaire	AC220V 1×200W 光源LED 光通量不小于24000lm 配吊杆 AC220V 1×200W LED luminaire The luminous flux is not less than 30000 lm with boom	套 Set	60		甲供 Supplied by Party A
2	双管荧光灯 Double tube fluorescent luminaire	AC220V 1×36W 光通量不小于3000lm AC220V 1×36W The luminous flux is not less than 3000 lm	套 Set	5		
3	壁灯(防水防尘型) Wall luminaire(waterproof and dustproof)	AC220V 1×32W 光通量不小于2500lm AC220V 1×32W The luminous flux is not less than 2500 lm	套 Set	4		
4	出口指示灯 Exit signs	光源LED(自带蓄电池,放电时间不低于120分钟) AC220V 3W 光通量不小于300lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) the luminous flux is not less than 300 lm	套 Set	1		
5	应急灯 Emergency luminaire	AC220V 2×8W 光通量不小于1600lm 带蓄电池,其放电时间不低于120分钟 AC220V 2×8W The luminous flux is not less than 1600 lm,the discharge time of battery is not less than 120min	套 Set	1		
6	方向指示灯 Emergency luminaire	光源LED(自带蓄电池,放电时间不低于120分钟) AC220V 3W 光通量不小于300lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) the luminous flux is not less than 300 lm	套 Set	2		
7	单联单控开关(防尘防水型) Single board single control switch (waterproof and dustproof)	AC250V 10A	只 Pcs	2		
8	屏蔽电缆 Shielded cable	ZRB-YJV22-P-0.6/1kV-3×4	米 Meter	1200		
9	屏蔽电缆 Shielded cable	NH-YJV22-P-0.6/1kV-3×4	米 Meter	500		用于备用照明回路 For Standby lighting

Description:

1. The power supply of high bay luminaires in pole 1 valve hall come from lighting box in control building,power supply cables are laid in cable bridge and electric power trunking,then cables protected by electric power trunking,then cables protected by electric power trunking 25×25 are leaded to high bay luminaires.
2. The power supply of high bay luminaires as stand-by lighting in pole 1 valve hall come from DC-AC shifter in control building,power supply cables are laid in cable bridge and electric power trunking,then cables protected by electric power trunking 25×25 are leaded to high bay luminaires.
3. The power supply of wall luminaires and fluorescent luminaires in pole 1 valve hall come from lighting box in control building,power supply cables are laid in cable bridge,then cables protected by hot galvanizing steel pipe φ25 are leaded to luminaires.
4. The power supply of exit signs and emergency luminaires in pole 1 valve hall come from lighting box in control building,power supply cables are laid in cable bridge,then cables protected by hot galvanizing steel pipe φ25 are leaded to luminaires.

巴基斯坦默蒂亚里—拉合尔直流输电工程
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±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - M C S - C S I - D D - T M - 0 5 0 2 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡初	2018.9.17
批准 Approved	[illegible]	2018.9.18

极2阀厅巡视走道层照明布置图
Lighting plan of Patrol aisle floor of pole 2 valve hall

比例 Scale: 1:200
图号 DWG NO.: 30-B003701S-D0502-004
版本 REV.: A

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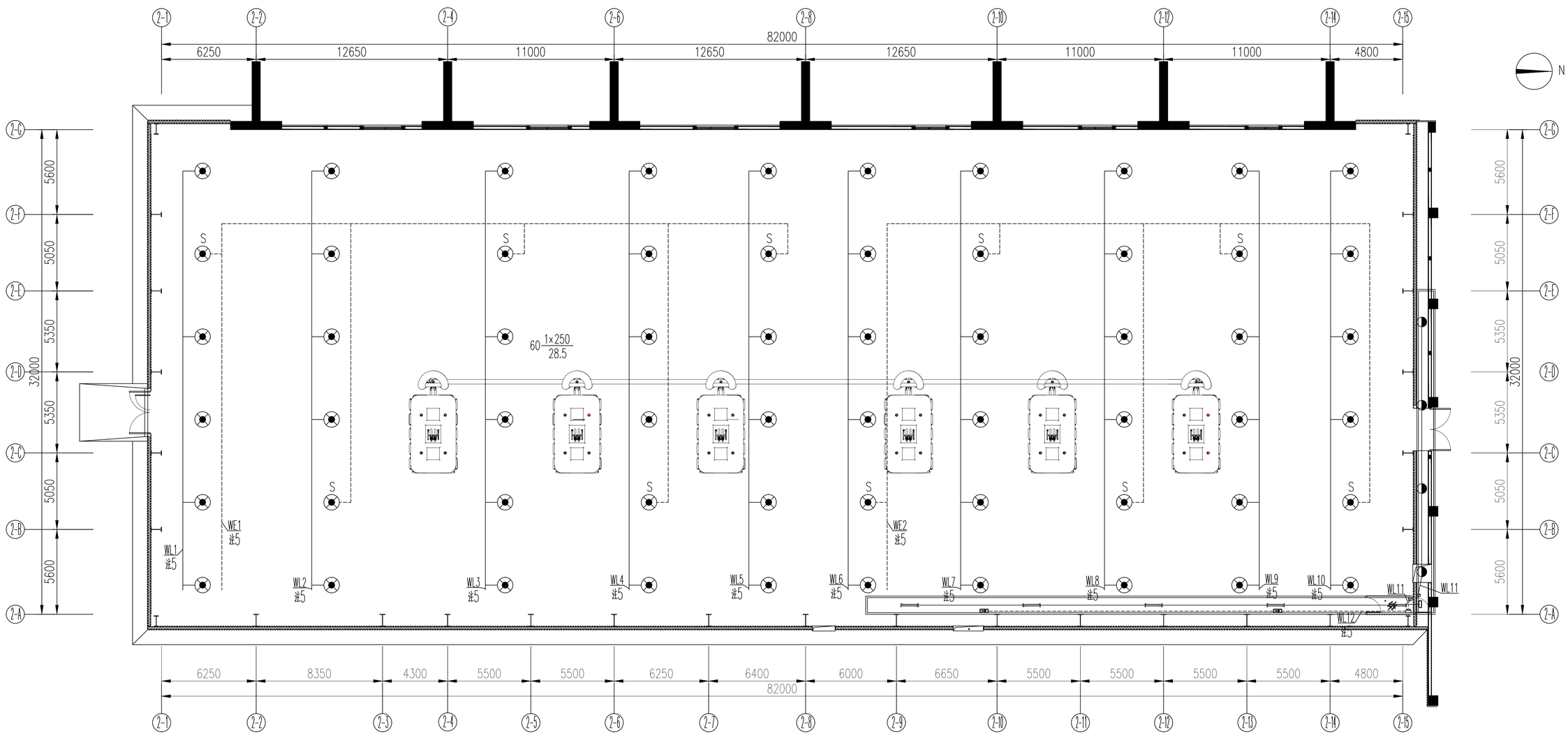
图形文件名



说明：

1. 极1阀厅正常照明高顶灯电源引自主控楼三楼照明箱，灯具线穿φ32热镀锌钢管明装敷设，至高顶灯较近位置处50×25线槽引至高顶灯，材料工总于材料工总表。
2. 极1阀厅备用照明高顶灯电源引自主控楼三楼事故照明箱，灯具线穿φ32热镀锌钢管明装敷设，至高顶灯较近位置处50×25线槽引至高顶灯，材料工总于材料工总表。
3. 极1阀厅壁灯和荧光灯电源引自主控楼三楼照明箱，灯具线穿φ32热镀锌钢管明装敷设引至灯具，材料工总于材料工总表。
4. 极1阀厅巡视走道出口指示灯和应急灯电源引自主控楼三楼照明箱，灯具线穿φ32热镀锌钢管明装敷设热镀锌钢管引至灯具，材料工总于材料工总表。
5. 本楼层灯具敷设原则采用穿管敷设，材料已工总于材料工总表，如施工方考虑便于施工等原因，可借用附近有余量的桥架等电缆通道。

材料表
Material list

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	图例 legend	备注 Notes
1	高顶灯 High bay luminaire	AC220V 1×200W 光源LED 光通量不小于24000lm 配吊杆 AC220V 1×200W LED luminaire The luminous flux is not less than 30000 lm with boom	套 Set	60		甲供 Supplied by Party A
2	双管荧光灯 Double tube fluorescent luminaire	AC220V 1×36W 光通量不小于3000lm AC220V 1×36W The luminous flux is not less than 3000 lm	套 Set	5		
3	壁灯(防水防尘型) Wall luminaire(waterproof and dustproof)	AC220V 1×32W 光通量不小于2500lm AC220V 1×32W The luminous flux is not less than 2500 lm	套 Set	4		
4	出口指示灯 Exit signs	光源LED(自带蓄电池,放电时间不低于120分钟) AC220V 3W 光通量不小于300lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) the luminous flux is not less than 300 lm	套 Set	1		
5	应急灯 Emergency luminaire	AC220V 2×8W 光通量不小于1600lm 带蓄电池,其放电时间不低于120分钟 AC220V 2×8W The luminous flux is not less than 1600 lm,the discharge time of battery is not less than 120min	套 Set	1		
6	方向指示灯 Emergency luminaire	光源LED(自带蓄电池,放电时间不低于120分钟) AC220V 3W 光通量不小于300lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) the luminous flux is not less than 300 lm	套 Set	2		
7	单联单控开关(防尘防水型) Single board single control switch (waterproof and dustproof)	AC250V 10A	只 Pcs	2		
8	屏蔽电缆 Shielded cable	ZRB-YJV22-P-0.6/1kV-3×4	米 Meter	1100		
9	屏蔽电缆 Shielded cable	NH-YJV22-P-0.6/1kV-3×4	米 Meter	400		用于备用照明回路 For Standby lighting

Description:

1. The power supply of high bay luminaires in pole 1 valve hall come from lighting box in control building,power supply cables are laid in cable bridge and electric power trunking,then cables protected by electric power power trunking,then cables protected by electric power trunking 25×25 are leaded to high bay luminaires.
2. The power supply of high bay luminaires as stand-by lighting in pole 1 valve hall come from DC-AC shifter in control building,power supply cables are laid in cable bridge and electric power trunking,then cables protected by electric power trunking 25×25 are leaded to high bay luminaires.
3. The power supply of wall luminaires and fluorescent luminaires in pole 1 valve hall come from lighting box in control building,power supply cables are laid in cable bridge,then cables protected by hot galvanizing steel pipe φ25 are leaded to luminaires.
4. The power supply of exit signs and emergency luminaires in pole 1 valve hall come from lighting box in control building,power supply cables are laid in cable bridge,then cables protected by hot galvanizing steel pipe φ25 are leaded to luminaires.

巴基斯坦默蒂亚里—拉合尔直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATC COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

默蒂亚里±660kV换流站
±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - M C S - C S I - D D - T M - 0 5 0 2 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡如	2018.9.17
批准 Approved	[illegible]	2018.9.18

极1阀厅巡视走道层照明布置图
Lighting plan of Patrol aisle floor of pole 1 valve hall

比例 Scale: 1:200
图号 DWG NO.: 30-B003701S-D0502-005
版本 REV.: A

420X594

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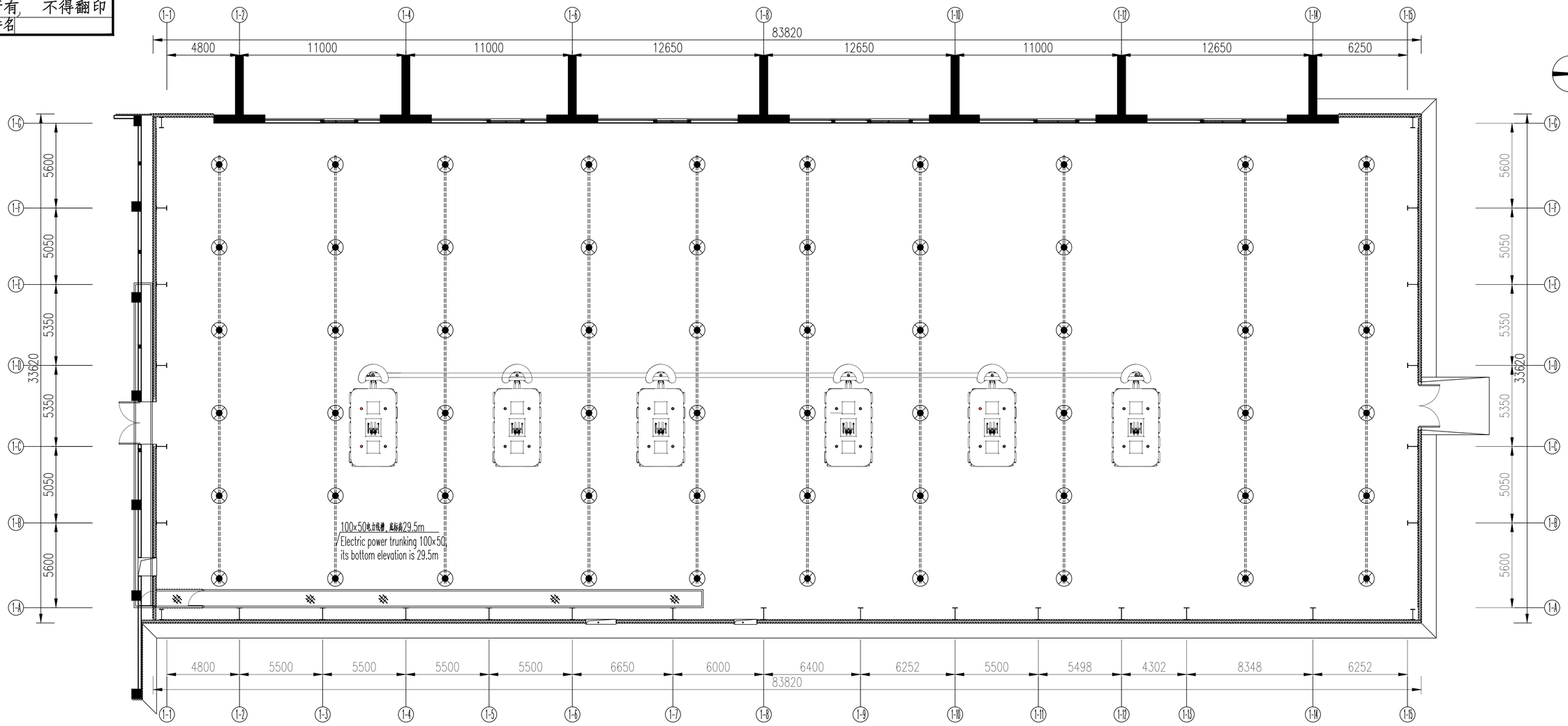
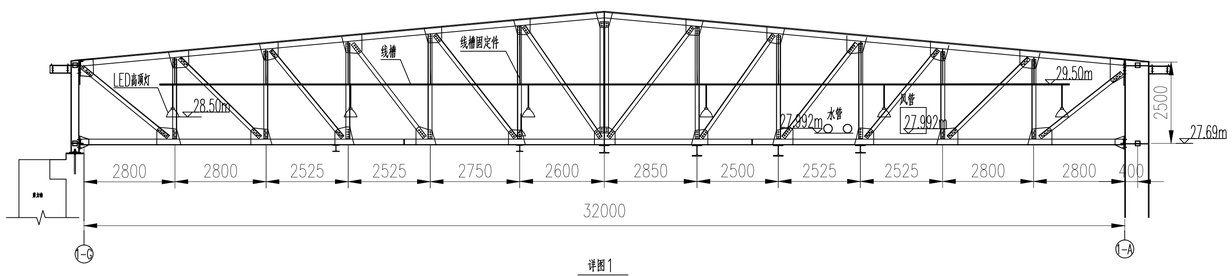
图形文件名





详图1
Detail1

材料表
Material list

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	图例 legend	备注 Notes
1	三通接头 Tee coupling	400×100变100×50 400×100 transform 100×50	米 Meter	10		
2	电力线槽 Electric power trunking	100×50(宽×高) 100×50(width×height)	米 Meter	320		
3	电力线槽固定件 Electric power trunking fastener		套 Set	120		
4	电力线槽 Electric power trunking	50×25	米 Meter	40		
5						
6						
7						
8						
9						

说明：

1. 100×50电力线槽悬吊固定于屋面檩条上，吊架长度现场放样制作，悬吊跨距2~3米，悬吊高度29.5米，见详图1。
2. 电力线槽采用35mm 铜绞线每10m与钢构件就近可靠连接，电力线槽间采用35mm²铜绞线跨接。热镀锌钢管采用φ20热镀锌圆钢与钢构件就近可靠连接。

Description:

1. Electric power trunking 100×50 is cantilevered on roof purlin, mading hanger on site, suspension span is 2~3m and the suspension height is 29.5m.
2. Stranded copper conductor 35mm ²is used to connect electric power trunking to nearby steel structure, stranded copper conductor 35mm ²is adopted as cross-connecting conductor between electric power trunkings,hot galvanizing round steel φ20 is used to connect hot galvanizing steel pipe to nearby steel structure.

巴基斯坦默蒂亚里—拉合尔直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

默蒂亚里±660kV换流站
±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - M C S - C S I - D D - T M - 0 5 0 2 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡初	2018.9.17
批准 Approved	[illegible]	2018.9.18

极2阀厅屋架线槽布置图
Trunking plan on roof truss of pole 2 valve hall

比例 Scale: 1:200
图号 DWG NO.: 30-B003701S-D0502-006
版本 REV.: A

420X594

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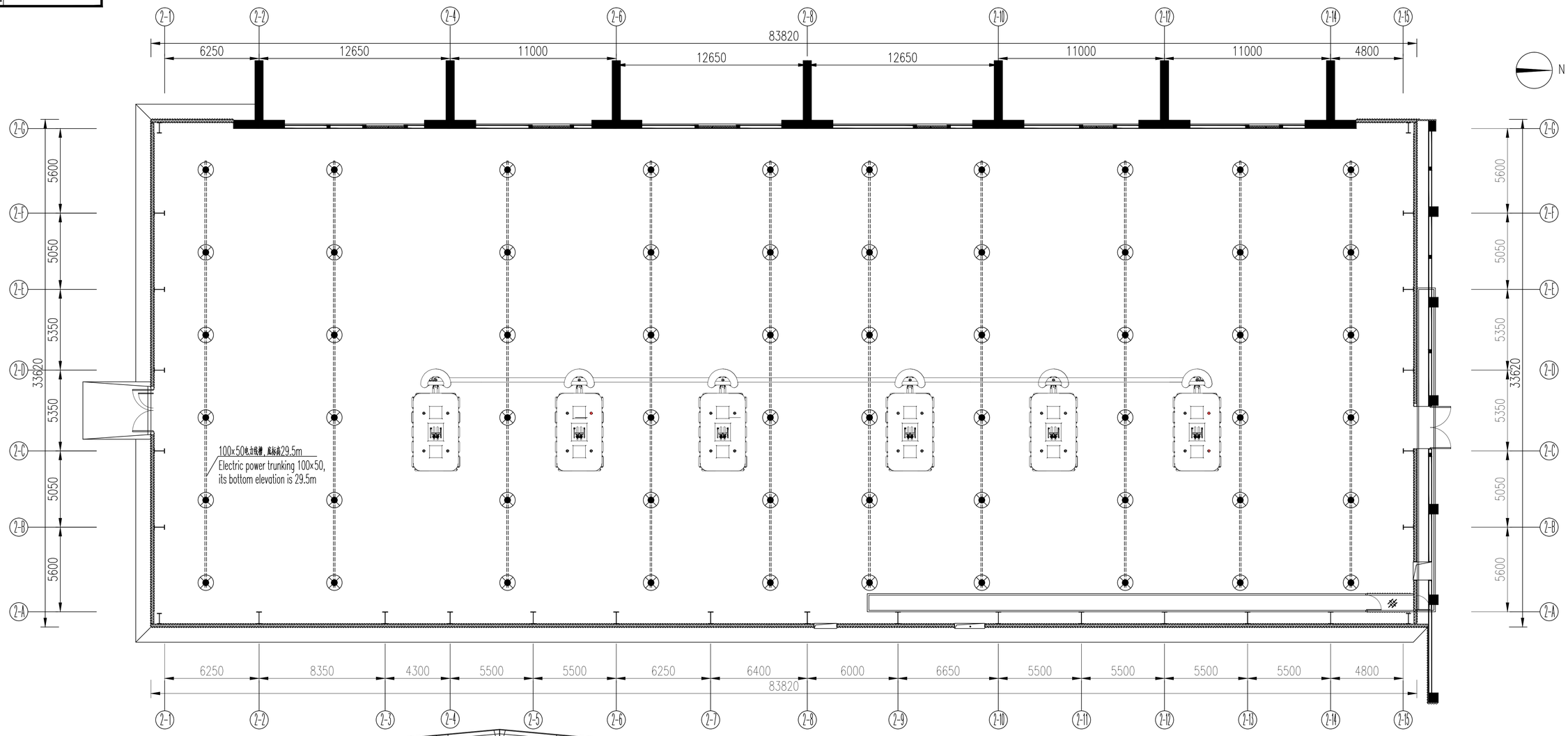
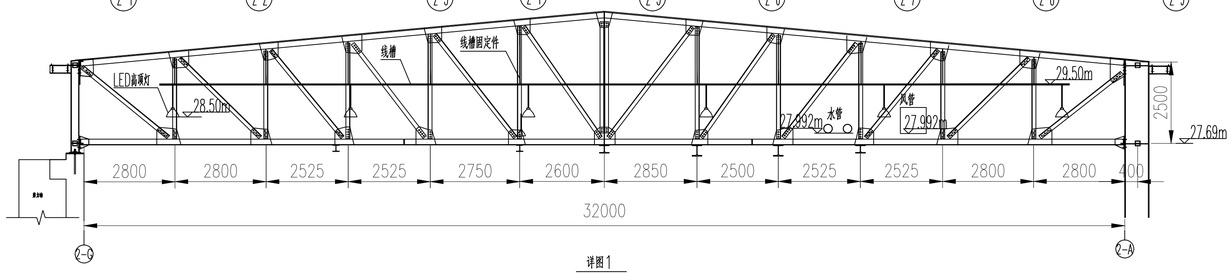
图形文件名





详图1
Detail1

材料表
Material list

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	图例 legend	备注 Notes
1	三通接头 Tee coupling	400×100变100×50 400×100 transform 100×50	米 Meter	10		
2	电力线槽 Electric power trunking	100×50(宽×高) 100×50(width×height)	米 Meter	320		
3	电力线槽固定件 Electric power trunking fastener		套 Set	120		
4	电力线槽 Electric power trunking	50×25	米 Meter	40		
5						
6						
7						
8						
9						

说明：

1. 100×50电力线槽悬吊固定于屋面檩条上，吊架长度现场放样制作，悬吊跨距2~3米，悬吊高度29.5米，见详图1。
2. 电力线槽采用35mm 铜绞线每10m与钢构件做可靠连接，电力线槽间采用35mm²铜绞线跨接。热镀锌钢管采用φ20热镀锌圆钢与钢构件做可靠连接。

Description:

1. Electric power trunking 100×50 is cantilevered on roof purlin, mading hanger on site, suspension span is 2~3m and the suspension height is 29.5m.
2. Stranded copper conductor 35mm ²is used to connect electric power trunking to nearby steel structure, stranded copper conductor 35mm ²is adopted as cross-connecting conductor between electric power trunkings,hot galvanizing round steel φ20 is used to connect hot galvanizing steel pipe to nearby steel structure.

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±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

默蒂亚里±660kV换流站
±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-MCS-CSI-DD-TM-0502-007A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡初	2018.9.17
批准 Approved	[illegible]	2018.9.18

极1阀厅屋架线槽布置图
Trunking plan on roof truss of pole 1 valve hall

比例 Scale: 1:200
图号 DWG NO.: 30-B003701S-D0502-007
版本 REV.: A

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图形文件名

J1FTZMX(J2FTZMX)
Pe=20.42kW(15.42kW)
Ijs=53.06A(30.32A)

T3N250 R100/3P

1#/2#电源进线
引自主控楼配电屏
from distribution panel in control building

详见后期电分册，由电气施工单位负责
It is shown in the power distribution volume

40*4热镀锌扁钢(暗敷)
与配电盒内PE端及配电盒金属件处可靠连接
Connecting the hot dip galvanized flat steel 40*4 (concealed laying) with metal case of distribution box and PE terminal block in distribution box

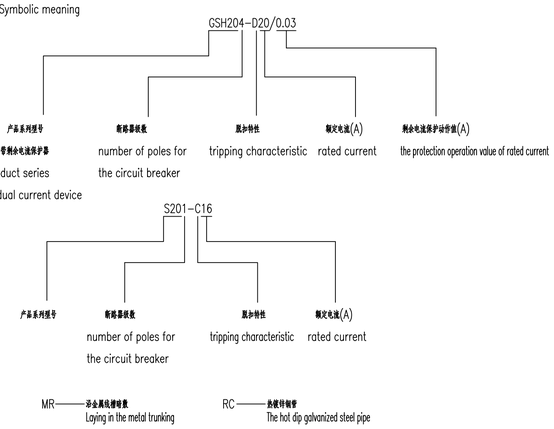
40*4热镀锌扁钢
Hot dip galvanized flat steel 40*4.

接地干线
Earthing bus conductor

母线载流量125A
Busbar carrying capacity is 125A

		Phase position 相位	Circuit number 回路编号	Capacity 容量(kW)	Current 电流(A)	Purpose 用途
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L1	WL1	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L2	WL2	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L3	WL3	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L1	WL4	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L2	WL5	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L3	WL6	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L1	WL7	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L2	WL8	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L3	WL9	1.25	7.58	Lighting 照明
S201-C16	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L1	WL10	1.25	7.58	Lighting 照明
S201-C10	ZRB-YJV22-P-0.6/1kV-3×4-RC32/MR	L2	WL11	0.308	1.87	Lighting 照明
S201-C10	NH-YJV22-P-0.6/1kV-3×4-RC32/MR	L3	WL12	0.044	0.273	Evacuation lighting 疏散照明
S202-C50	NH-YJV22-0.6/1kV-2×10	L3	WL13			Standby 备用
S203-C20		L1L2L3				Standby 备用
GSH202-C20/0.03		L1				Standby 备用
S201-C20		L2				Standby 备用

说明：符号含义
Description:Symbolic meaning



极1阀厅照明箱J1FTZMX/极2阀厅照明箱J2FTZMX配置接线图
Distribution diagram of lighting box J1FTZMX for pole 1 valve hall /J2FTZMX of pole 2 valve hall

由EPS屏引接
To in control building

INT 50/50A2P

母线载流量30A
Busbar carrying capacity is 50A

		Phase position 相位	Circuit number 回路编号	Capacity 容量(kW)	Current 电流(A)	Purpose 用途
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE1	1.25	7.58	Standby lighting 1 of pole 1 valve hall 极1阀厅备用照明1
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE2	1.25	7.58	Standby lighting 2 of pole 1 valve hall 极1阀厅备用照明2
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE3			Standby 备用
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE4			Standby 备用
S202-C20			WE5			Standby 备用

由EPS屏引接
To in control building

INT 50/50A2P

母线载流量30A
Busbar carrying capacity is 50A

		Phase position 相位	Circuit number 回路编号	Capacity 容量(kW)	Current 电流(A)	Purpose 用途
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE1			Standby 备用
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE2			Standby 备用
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE3	1.25	7.58	Standby lighting 1 of pole 2 valve hall 极2阀厅备用照明1
S202-C16	NH-YJV22-P-0.6/1kV-3×4/MR		WE4	1.25	7.58	Standby lighting 2 of pole 2 valve hall 极2阀厅备用照明2
S202-C20			WE5			Standby 备用

阀厅事故照明箱1/2配置接线图
Distribution diagram of DC-AC shifter FTQHX valve hall

说明：
1. 极1阀厅照明箱、极2阀厅照明箱及事故照明箱均位于主控楼三楼。

Description:
1. Lighting box J1FTZMX for pole 1 valve hall,Lighting box J1FTZMX for pole 2 valve hall and DC-AC shifter FTQHX are all installed in Third floor of control building.

巴基斯坦默蒂亚里—拉合尔直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR
国家电网 STATE GRID
中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR
East China Electric Power Design Institute Co.,Ltd.

默蒂亚里±660kV换流站
±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-MCS-CSI-DD-TM-0502-008A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡如	2018.9.17
批准 Approved	[illegible]	2018.9.18

照明箱配置接线图
Distribution diagram of lighting box

比例 Scale	1:200	图号 DWG NO.	30-B003701S-D0502-008	版本 REV.	A

420X594

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图形文件名

材料表
Material list

序号 Item No.	名称 Name	型号及规格 Model and specification	单位 Unit	数量 Quantity	备注 Remarks
1	照明箱 Lighting distribution box	600×600×200(高×宽×深) 600×600×200(height×width×depth)	个 Pcs	2	
2	事故照明箱 DC-AC shifter	600×600×200(高×宽×深) 600×600×200(height×width×depth)	个 Pcs	2	
3	吸顶灯(防水防尘型) Ceiling luminaire(waterproof and dustproof)	AC220V 1×32W 荧光灯光源(不含紫外光) 光通量不小于2500lm AC220V 1×32W fluorescent luminaire(without ultraviolet light) The luminous flux is not less than 4500 lm	套 Set	2	
4	高顶灯 High bay luminaire	AC220V 1×200W 光源LED(不含紫外光) 光通量不小于24000lm 配吊杆 AC220V 1×200W LED luminaire(without ultraviolet light) The luminous flux is not less than 24000lm with boom	套 Set	120	甲供 Party A provides
5	壁灯(防水防尘型) Wall luminaire(waterproof and dustproof)	AC220V 1×32W 荧光灯光源(不含紫外光) 光通量不小于2500lm AC220V 1×32W fluorescent luminaire(without ultraviolet light) The luminous flux is not less than 2500 lm	套 Set	8	
6	出口指示灯 Exit signs	光源LED(自带蓄电池,放电时间不低于120分钟) 不含紫外光 AC220V 3W 光通量不小于300lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) without ultraviolet light the luminous flux is not less than 300 lm	套 Set	6	
7	应急灯 Emergency luminaire	光源LED(自带蓄电池,放电时间不低于120分钟) 不含紫外光 AC220V 2×8W 光通量不小于1600lm luminaire source:LED(Built-in battery whose discharge time of battery is not less than 120min) without ultraviolet light the luminous flux is not less than 1600 lm	套 Set	4	
8	单联单控开关(防尘防水型) Single board single control switch (waterproof and dustproof)	250V 10A	只 Pcs	6	
9	屏蔽电缆 Shielded cable	ZRB-YJV22-P-0.6/1kV-3×4	米 Meter	2800	甲供 Party A provides
10	屏蔽电缆 Shielded cable	NH-YJV22-P-0.6/1kV-3×4	米 Meter	1200	
11	电力线槽 Electric power trunking	100×50(宽×高) 100×50(width×height)	米 Meter	700	
12	电力线槽 Electric power trunking	50×25(宽×高) 50×25(width×height)	米 Meter	150	
13	三通接头 Tee coupling	400×100变100×50 400×100 transform 100×50	个 Pcs	20	
14	热镀锌钢管 Hot galvanizing steel pipe	φ50	米 Meter	60	
15	热镀锌钢管 Hot galvanizing steel pipe	φ32	米 Meter	1900	
16	电力线槽固定件 Electric power trunking fastener		套 Set	120	
17	铜绞线 Stranded copper conductor	35mm²	米 Meter	220	
18	铜鼻子 copper cable lug	用于35mm²铜绞线 for 35mm² stranded copper conductor	个 Pcs	450	
19	热镀锌圆钢 Hot galvanizing round steel	φ20	米 Meter	30	
20	双管荧光灯 Double tube fluorescent luminaire	AC220V 1×36W 光通量不小于3000lm AC220V 1×36W The luminous flux is not less than 3000 lm	套 Set	10	

说明：

1. 表中所列材料量为极1极2两个阀厅所需材料量。

Description:

1. The materials in the list contains materials required for pole 1 valve hall and pole 2 valve hall.

巴基斯坦默蒂亚里–拉合尔直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

默蒂亚里±660kV换流站
±660kV Matiari Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-MCS-CSI-DD-TM-0502-009A

	姓名 Name	日期 Date
设计 Designed	林新明	2018.9.10
校核 Checked	李旭	2018.9.11
审核 Reviewed	金怡初	2018.9.17
批准 Approved	[illegible]	2018.9.18

材料表
Material list

比例 Scale | 图号 DWG NO. 30-B003701S-D0502-009 | 版本 REV. A

297X420